Lizard Jaw Auto-Segmentation Report

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1 Project Description

The purpose of this project is to automate the segmentation of teeth and lower jaws from 3D Micro CT scans of Anolis lizards. The current state of the art for segmenting these teeth is to open the scan file in a program such as 3D Slicer and manually draw the segments on each frame, which can take up to 2 hours per lizard scan to be segmented. This presents a major barrier to collecting a large number of samples for comparative analysis. By automating the segmentation process, we can produce a large number of segmented examples in a fraction of the time, enabling inter- and intra-species comparison studies.

2 Challenges

This project presents several main challenges. First, the lizard bodies are positioned differently in each scan. this means that getting a high quality view of the lower jaw and teeth in each scan is difficult. Second, 3D segmentation is a challenging and computationally intensive problem, and the state of the art methods are not well suited to this project. Finally, the final use case of these segmentations, the Dental Dynamics slicer module, requires that each tooth be segmented individually, so we need to find different segmentation masks for each individual tooth rather than the teeth as a whole.

3 Technical Design and Alternatives

The proposed solution to this problem is broken down into three phases:

- Preprocess full-body lizard CT scan to cropped head which is aligned with slicing axis
- Segment the lower jaw and teeth from each slice of the head scan
- Combine the individual slices into a cohesive 3D model of the lower jaw

3.1 Phase One: Preprocess Scan

Preprocessing the scan to retrieve a clean view of the anatomy of interest makes the downstream segmentation task much clearer. In order to do this we must automatically determine the 3D transformation which will crop and align the head with the xyz axis of slicing. This transformation will be different for every file and can be represented as a 3D affine transformation matrix M.[1] Using image registration techniques, we can take an example head H_{source} which has been manually set to the ideal alignment and find the transformation M which moves it to match the head H_{target} in the full body scan in question (i.e. $H_{target} = MH_{source}$). Thus the inverse of this transformation, M^{-1}, will satisfy $H_{source} = M^{-1}H_{target}$, meaning that it is the transformation from head positioned in the full body scan to the ideally aligned head.

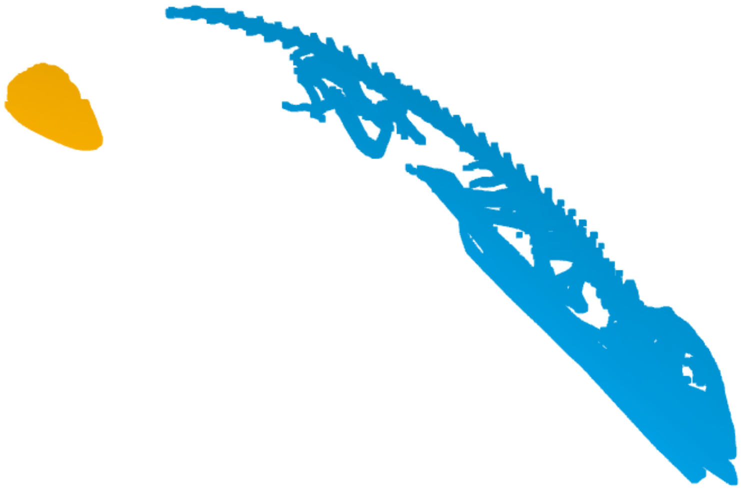

Figure 1: H_{source} (yellow) and the target model (blue) before registration process

3.1.1 Global Registration

The first step in achieving an accurate transformation is global registration, which is expected to roughly align H_{source} with the head area of the target body. Performing registration on a full volume, which has roughly 5e8 pixels in it, is expensive. To reduce computation while achieving an accurate mapping, it is then necessary to reduce the amount of information in the volumes by thresholding away the bottom half of pixel intensities which removes the soft tissue from the volume, leaving behind the lizard skeleton. This is then converted to a point cloud and downsampled aggressively to a voxel size of 15 by 15 by 15 pixels.

RANSAC The downsampled point clouds of the aligned head and full body are then used for the Random Sample Consensus (RANSAC) global registration algorithm. RANSAC works by iteratively selecting random samples of both point clouds, and calculating a transformation that would make them align. The algorithm then tests this proposed transformation by determining how well it fits with both whole models. If the proposed sample has a sufficiently high proportion of inliers (points which fit the transformation) or the algorithm reaches the maximum number of iterations it terminates. The output of the algorithm is our global transformation matrix M_{global}.

Fast Global Registration An alternative approach explored was replacing the RANSAC algorithm with Fast Global Registration, an algorithm proposed by Zhou et al. that uses pre-computed feature correspondences to achieve accurate global registration much more quickly than traditional methods.[2] Fast Global Registration has several attractive qualities, from runtime to stochastic selection process. Experimentally, I found that Fast Global Registration led to worse results than RANSAC, so I chose not to use it.

3.1.2 Coarse Cropping

In order to produce the cleanest match in the local refinement step, it is important to remove extraneous information from the full body scan. To do this, I created a coaxial bounding box twice the size of H_{source}, and cropped the full body to only the part of the model which fell within that area.

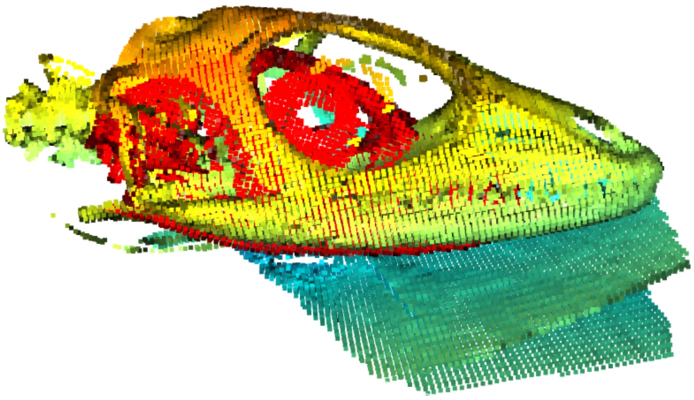

Figure 2: H_{source} (red) and the target model (yellow-orange) after global registration and coarse cropping

3.1.3 Local Registration

In order to achieve the closest match possible with the next step of the process, we resample both point clouds to a more fine-grained voxel size of 3 by 3 by 3 pixels. We then use the Iterative Closest Point (ICP) algorithm with included scaling to achieve the closest possible match between H_{source} and H_{target}.

3.1.4 Fine Cropping

Finally, I created a coaxial bounding box 10% larger than H_{source} and cropped the target model to this area, obtaining the final H_{target}.

3.1.5 Final Transformation

Now that we have calculated M_{global} and M_{local}, we can determine our final transformation matrix M and apply it to our original volume.

$$M = M_{local} M_{global}$$

Therefore:

$$M^{-1} = M_{global}^{-1} M_{local}^{-1}$$

3.2 Phase Two: Segment Teeth and Jaw from Head Images

In order to segment the lower jaw and teeth, the 3d scan of the head is converted to a stack of individual frames, where each frame is fed through a Vision Transformer to extract segmentation masks and classifications for each structure in the image. This is divided into four possible classes: "background", "lower jaw", "tooth", and "other bone".

For these purposes, only the teeth on the lower jaw are considered part of the "tooth" class, upper teeth should be labeled as "other bone".

As the outputs of this pipeline are 3D segmentation masks, the model outputs are what we will be working with for the remaining steps.

3.2.1 Model Training and Selection

For this project, it is important to achieve high quality segmentations, but also to be able to extract information about which structures the masks correspond to. For this reason I decided to use a model capable of performing panoptic segmentation, which is simultaneous segmentation and classification of an entire image. The current state of the art in many computer vision tasks is achieved by various transformer models. A recent paper from Meta Research proposed the Masked-Attention Mask Transformer (Mask2Former), which is a vision transformer model designed to be capable of semantic, instance, and panoptic segmentation.[3] The Mask2Former model used is an implementation through the popular machine learning python library HuggingFace, and includes pre- and post-processing functions.

3.2.2 Data Preprocessing

Before using the Mask2Former model, the 3D head model is separated into individual slices showing the sagittal plane. Each slice follows a multistep preprocessing pipeline consisting of:

- Resizing the frame to a standard size
- Converting the image to Color
- Randomly selecting images to flip horizontally

For model training and validation data, the corresponding ground truth masks undergo the same preprocessing pipeline, and are then processed to collect a dictionary of information about each of the individual segments in the frame. During this process, each segment is assigned a class as described above. This information is fed into the model as the ground truth, so that the model learns to generate the individual segments.

3.3 Phase Three: Combine Individual Frames into Whole Volume

The main challenge with combining frames into a single volume is that the set of segments output by the Mask2Former model is not in any particular order. Segments of the same tooth on two different frames can have different segment ids, and the only way to determine which segments correspond is to consider their locations. As a consequence of aligning the skull to the coordinate axes and segmenting in the sagittal view, segments of the same tooth are roughly aligned with each other along the z-axis.

The method I use to combine frames is as follows: For each tooth t in the current frame, find the set S of all segments on the next frame that overlap with t. Next compute the Intersection over Union (IoU) score for each segment in S. If the highest scoring segment is any class other than tooth, take no action. If the highest scoring segment is a tooth, change that tooth's ID to match the ID of t. Repeat this process for each frame.

Alternative Approaches One alternative approach to solving this problem is to treat each successive frame in the volume as a video, using object detection and tracking methods to associate items between frames. This is a solution that I have not had time to pursue, but I believe that this or some other alternative deserves consideration.

4 Results

For the purposes of this project and the point clouds involved, I have found that an inlier RMSE (RMSE for only the points considered inliers) of < 6 for the local registration step is sufficient to denote a close match that can be used for segmentation. Due to the downsampling and cropping steps, it is common for sufficient and insufficient solutions to achieve a perfect fitness score (proportion of points in the model which are considered inliers), so this metric is not helpful for measuring registration performance. The current pipeline has achieved sufficiently low RMSE on each tested example, a set of 4 randomly selected models.

Model performance In order to measure model performance, we use both loss as well as panoptic quality, a metric which balances segmentation and classification performance into a single score.

$$PQ = SQ * RQ$$

Where SQ is segmentation quality, a metric of how well the proposed segmentations match the ground truth, and RQ is recognition quality, a metric of how well the classifications produced by the model match the ground truth classes for each segment. Panoptic quality can be calculated for the dataset as a whole, or broken down to show the model performance on each class. It is calculated on a scale of $[0, 1]$, with greater values being better.

The segmentation model I trained achieved a final panoptic quality of 0.70, with segmentation quality of 0.798 and recognition quality of 0.87. These numbers are boosted by high SQ and RQ scores on other bone and background classes. The most important classes for this project, tooth and lower jaw, show poor performance on the validation dataset. There is room for improvement through further model training, as well as adjustments to model architecture or hyperparameter values.

5 Results Visualization

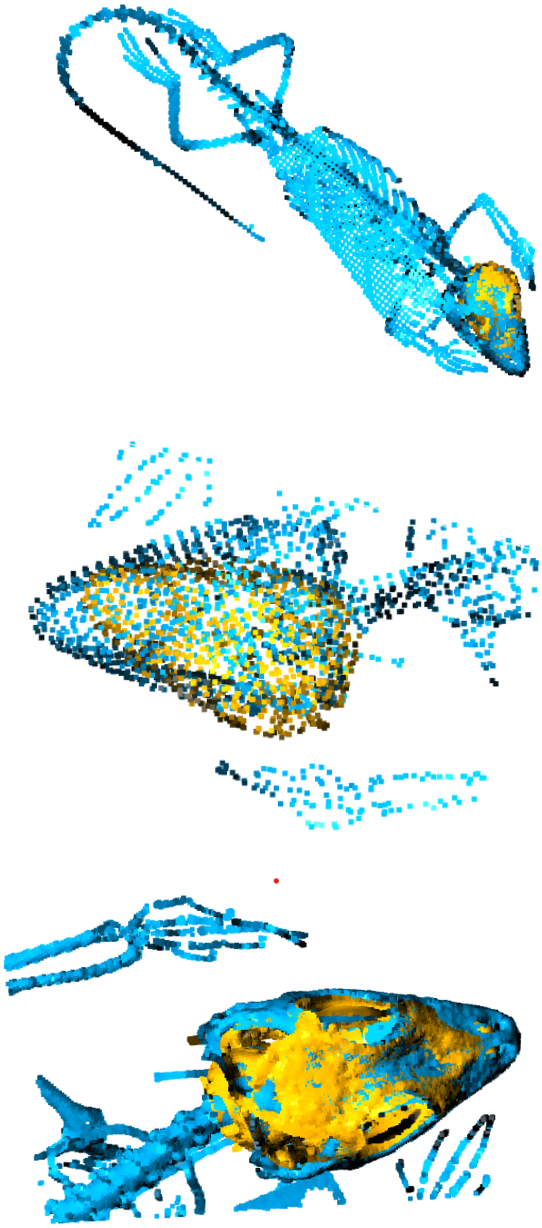

Figure 3: Step-by-step results for the 3D registration pipeline. In order: global registration, coarse cropping, and local registration

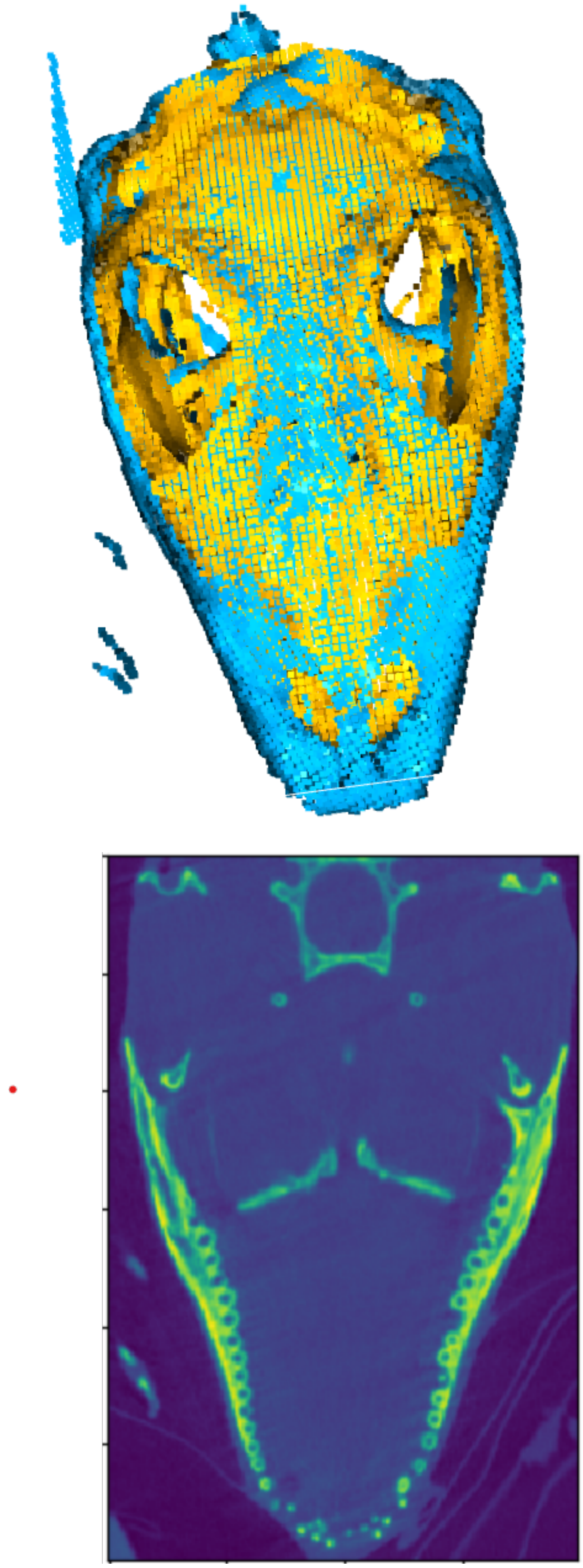

Figure 4: Step by step results for 3D registration continued. Fine cropping and final view of pixels after transformation.

6 Remaining Work

The largest area of remaining work in this project is improving the segmentation model performance. Additional training data or changes in model architecture, optimizers, or metrics are all potential improvements. In addition, the postprocessing step to combine adjacent frames into a single volume needs validation and refinement.

I have begun research into how to deploy the final pipeline for use by Stroud lab and other researchers. The first option is to create a standard python script or docker container which can be run locally. This has the advantage of using the existing code as written, but requires end users to have familiarity with configuring a development environment or a docker host. The second option is to refactor the code into a module which can be imported through 3D Slicer. This would make it much more usable for researchers, as well as allowing the pipeline to use built-in slicer modules such as transformations and file I/O.

References

[1] S. Beauchemin. "Three-Dimensional Affine Transformations." (2020), [Online]. Available: https://www.csd.uwo.ca/~sbeauche/CS3388/CS3388-Affine-Transformations.pdf (visited on 11/20/2024).

[2] Q.-Y. Zhou, J. Park, and V. Koltun, "Fast global registration," in *Computer Vision – ECCV 2016*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds., Cham: Springer International Publishing, 2016, pp. 766–782, ISBN: 978-3-319-46475-6.

[3] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 1290–1299.